Chinese Twin Agricultural Tariffs Impact on Dairy Supply

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April 13, 2026

Acknowledgements

The authors would like to thank Prof. Bryan Lohmar from College of Agriculture, Food & Environmental Sciences, Cal Poly; Prof. Conner Mullally, Prof. Christa Court, and Prof. Xiaohui Qiao from Department of Food Resource Economics, University of Florida for their valuable feedback on earlier drafts of this manuscript. This research was not supported by any external funding. We also extend our gratitude to the attendees of the 2025 Agricultural & Applied Economics Association (AAEA) and Western Agricultural Economics Association (WAEA) Joint Annual Meeting for their insightful comments. This paper accepted by the 2026 World Congress Environment & Resource Economics (WCERE) for paper presentation. All remaining errors are our own.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Highlights:

- Develops a “twin tariffs” framework to analyze simultaneous tariffs on upstream feed inputs and downstream dairy markets.

- **Asymmetric Import Elasticity:** China’s alfalfa hay imports are inelastic, while dairy imports are more substitutable across foreign suppliers.

- **Price and Cost Impact:** China’s domestic raw milk price rose by about 0.5% relative to the pre-tariff period, mainly through the feed-cost channel.

- **Compounding Welfare Losses:** Welfare simulations demonstrate that joint "twin tariffs" generate significantly larger economic losses than the sum of individual tariffs applied in isolation.

- **Policy Mitigation:** Results suggest that domestic support targeted specifically at upstream feed constraints can substantially reduce the welfare losses caused by trade-war-induced supply chain disruptions.

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Abstract

This paper develops a "Twin Tariffs" theory, in which China imposes tariffs on feed inputs and dairy products. The retaliatory tariffs on an upstream feed input (alfalfa hay) and a downstream output (dairy products) jointly affect import adjustments and the costs of domestic dairy products in China. Tariff increases are associated with lower Chinese imports from the United States, with stronger adjustments in the alfalfa-related channel than in the dairy-output channel. We use a time-series growth response model with fixed effects to estimate dynamic elasticities and lagged pass-through from tariff changes to import quantities and domestic farm-gate milk prices, and a large-country partial-equilibrium twin tariff framework to interpret terms-of-trade and welfare implications under alternative tariff and subsidy scenarios. Chinese domestic milk product prices rose by 0.5% during the trade war period (2018-22) due to the higher cost of alfalfa and alternative dairy imports. Welfare simulations indicate that the joint tariff regime generates larger losses than either tariff alone, while domestic support targeted at the upstream feed constraint can substantially reduce those losses. Overall, the combination of alfalfa and dairy tariffs results in welfare loss.

Keywords: Tariffs; Alfalfa; Dairy; Trade War; Vertical Supply Chain
JEL Codes: F13, Q17, Q18

1 Introduction

Trade barriers distort resource allocation, impacting producer viability, consumer welfare, and broader food security (Interesse, 2024; Fajgelbaum et al., 2024; Staples and McCullough, 2025). Yet, the majority of the agricultural trade literature evaluates these disruptions exclusively through the lens of single-commodity incidence: a tariff raises domestic prices, reduces imports, and generates deadweight loss. What remains largely unexplored is how trade barriers compound when applied across vertically linked supply chains. The 2018 U.S.-China trade war provides a prime, unexamined case study of a policy shock that we term "twin tariffs" —the simultaneous imposition of duties on both upstream inputs and downstream products. China enacted retaliatory tariffs of more than 25% on agricultural goods from the U.S., which affected 86.3% of HS 6-digit products in 2018 due to the trade war (Fajgelbaum and Khandelwal, 2022; Qian, 2023; Xu et al., 2025). Specifically, these retaliatory tariffs targeted U.S. alfalfa hay (a critical feed input) alongside processed dairy products (the final output). This dual-targeting created cost pressures and cross-market distortions that conventional, single-commodity analyses fundamentally fail to capture.

The magnitude of these cross-market distortions is dictated by a structural asymmetry in China's agricultural imports. On the input side, China lacks a comparative advantage in land-intensive forage crops, rendering its domestic dairy sector highly vulnerable to feed supply shocks (Hou et al., 2024). By 2023, the United States accounted for 89.9% of China's alfalfa hay imports, which alone supplied approximately half of the nation's total demand (Hou et al., 2024; Wan et al., 2022). Because Chinese producers cannot easily substitute this high-quality feed with alternative global suppliers, their import demand is inelastic, forcing them to absorb the brunt of tariff-induced cost increases. Conversely, on the output side, China's dairy imports are sourced from a more diversified pool of suppliers—including New Zealand, the European Union, and Australia. This diversification provides dairy products buyers with substantial substitution flexibility, making their demand more elastic than alfalfa. Consequently, domestic dairy farmers are caught in a classic economic squeeze: they face

unavoidable, compounded cost increases for critical inputs, yet they are severely constrained in their ability to pass these costs onto consumers who can easily pivot to alternative imported dairy products (Fertő et al., 2021; Koppenberg and Hirsch, 2022). This stark asymmetry—inelastic input demand coupled with elastic output demand—is the central mechanism dictating how "twin tariffs" [1] amplify deadweight losses and disproportionately redistribute the welfare burden onto domestic producers.

Motivated by the failure of single-commodity models to capture these compounding distortions and structural asymmetry in the twin-tariff case, this paper asks: 1) how do simultaneous tariffs on an upstream input and a downstream output propagate through a domestic agricultural sector; 2) what are the net welfare consequences of this twin tariff regime? Specifically, we seek to quantify the cross-market pass-through of feed tariffs to domestic farm-gate milk prices and decompose the resulting welfare losses.

To answer these questions, we develop a formal two-market partial equilibrium model with an explicit vertical cost linkage, deriving propositions about cross-market tariff pass-through, asymmetric import substitution, and the welfare amplification unique to twin tariffs. We bring the model to the data using monthly bilateral trade and price series from 2005 to 2025. Empirically, we estimate distributed-lag reduced-form specifications to trace tariff transmission to import quantities and farm-gate milk prices, supplement these estimates with interaction and mechanism checks, and then use the estimated relationships to motivate counterfactual welfare scenarios under alternative policy regimes.

Our empirical results provide evidence consistent with the theoretical mechanism, but the strength of that evidence differs across specifications. In the reduced-form estimates, increases in the alfalfa tariff are positively associated with subsequent increases in China's farm-gate milk price, which is consistent with cost pass-through from imported feed to domestic dairy production. By contrast, the dairy tariff does not generate a comparably robust positive domestic price effect, which is consistent with greater substitution toward non-U.S. dairy

[1] Twin tariff: tariffs imposed on both inputs (alfalfa) and outputs (dairy), tariffs imposed on alfalfa increased the dairy inputs cost, then the dairy supply curve shifts to the left.

suppliers. Together, these findings support the paper's central argument that upstream input tariffs can matter more for domestic dairy-price pressure than downstream output tariffs in a setting with asymmetric substitution possibilities.

Translating these price effects into aggregate welfare (Villoria, 2026), we find that the twin tariff regime generates a net economic loss of $15.7 billion. Crucially, this total far exceeds the sum of the deadweight losses that would occur if each tariff were imposed in isolation. Our structural model reveals that this excess loss is entirely driven by the cross-market distortion: the upstream alfalfa tariff shifts the downstream dairy supply curve inward, amplifying the deadweight loss generated by the direct dairy tariff.

Given these significant compounding losses, we evaluate the efficacy of domestic policy responses to supply chain shocks. We find that domestic alfalfa production subsidies successfully offset the input-cost channel, reducing the net welfare loss of the twin tariff regime by over 80%. This confirms that targeting the constrained, inelastic upstream input is substantially more efficient than providing equivalent protection to the downstream output.

By quantifying these mechanisms, this analysis adds concrete empirical and theoretical value to the broader literature on trade policy. Existing research on the U.S.–China trade war extensively documents tariff pass-through to import prices (Amiti et al., 2019), bilateral trade reductions (Fajgelbaum et al., 2024), and sector-specific welfare losses (Chor and Li, 2024). But within the agricultural sector, no prior work has formally analyzed the interaction between input and output tariffs within a vertically linked agricultural supply chain, or derived the welfare implications of this twin tariff mechanism. While the theoretical trade literature on tariff escalation (Smith and Glauber, 2020; Grant et al., 2021; Richman et al., 2014) and input–output tariff structures provides conceptual foundations, it has not been applied to the case of feed–dairy supply chains in a large-country setting (Fertő et al., 2021). By formalizing and estimating the welfare amplification unique to twin tariffs, this paper provides actionable guidance for evaluating tariff policy in vertically integrated agricultural markets.

2 Background and Conceptual Framework

2.1 Dairy and alfalfa trade

China's dairy sector has experienced steady per capita consumption growth, driven by urbanization, rising incomes, and government nutrition campaigns emphasizing the role of protein and calcium intake (Gooch et al., 2017). While per capita consumption remains on a steady upward trajectory, domestic production has struggled to keep pace. Output growth is significantly hindered by land constraints and escalating feed costs, particularly within the "Golden Milk Source Belt" of Inner Mongolia, where accounts for about 20% of China's domestic milk production (see appendix A1).

The gap between domestic supply and demand has made China the second-largest milk importer worldwide. In 2024, China imported $1.42 billion worth of milk and became the second largest importer of milk in the world. The United States serves as the third largest supplier of dairy products to China, primarily providing whey, milk powder, and lactose. These are all primary processed milk products. While New Zealand and the European Union remain the primary sources of China's liquid milk imports, the U.S. dominates the feed market; it accounted for 89.9% of China's alfalfa hay imports in 2023(Hou et al., 2024).

Alfalfa is the third most valuable cash crop in the U.S., with an estimated economic value of $8.7 billion in 2023 (U.S. Department of Agriculture, Economic Research Service (USDA ERS), 2024). It is also recognized as one of the most important forage crops globally due to its high nutritional value for livestock, making it a vital export commodity (Cantor et al., 2022).

2.2 The "twin tariff' intervention

The 2018 U.S.-China trade conflict introduced a distinct policy phenomenon defined here as "twin tariffs": the simultaneous imposition of retaliatory tariffs on upstream production inputs (forage) and downstream final products (dairy) (USDA FAS, 2023a). This dual-layered

intervention has evolved through several distinct phases, creating a unique environment for analyzing supply chain shocks.

Phase I: Initial Escalation (2018–2019): The conflict began in July 2018, when China imposed an initial 25% retaliatory tariff on U.S. agricultural products, including alfalfa and a broad range of dairy products (Xu et al., 2025; USDA ERS, 2024). By 2019, the effective tariff burden intensified; cheese products faced total duties of 33–37%, representing a combination of Most-Favored-Nation (MFN) rates and Section 301 retaliation. Subsequent rounds in September 2019 pushed rates for fresh and grated cheese to 38–42%, while fermented milk and buttermilk reached levels as high as 50%.

Phase II: Managed Retaliation and Exclusions (2020–Early 2025): Following the Phase One Agreement in February 2020, the policy landscape shifted from blanket escalation to a regime of "managed retaliation." While retaliatory tariffs were not fully rescinded, China introduced a robust exclusion process that bifurcated the impact on the supply chain: 1) upstream product: alfalfa was granted exclusion status as early as September 2019, while whey intended for feed use received repeated extensions through February 28, 2025. 2) downstream products: Section 301 tariffs remained active for most final dairy products despite expanded market access for U.S. facilities. Exclusions in this category were granted only sparingly on a case-by-case basis, primarily to alleviate acute cost pressures within the domestic dairy-feed system.

Phase III: The 2025 Escalation and Geneva Negotiations: The trade environment experienced renewed volatility in early 2025, with tariffs escalating significantly. However, following the Geneva negotiations in May 2025, China implemented a temporary 90-day reduction of the additional retaliatory duty—dropping it from 125% to 10%. Despite this recent reduction, earlier tariff layers and product-specific measures persisted, resulting in a heterogeneous effective tariff burden across different agricultural categories.

2.3 China's diversification and mitigation policy

To mitigate cost shocks and stabilize farm-gate milk prices, China introduced producer subsidies for local forage growers and expanded support for regional dairy production. Beyond domestic subsidies, China has actively pursued trade diversification to reduce its reliance on U.S. China has systematically sought alternative markets for its agricultural needs (Qian, 2023; Rose, 2024). This strategy is further evidenced by China's ratification of the Regional Comprehensive Economic Partnership (RCEP)—the world's largest free trade area—and its application to the Comprehensive and Progressive Trans-Pacific Partnership (CPTPP) (Rose, 2024; Li et al., 2020).

These diversification efforts are designed to diminish the impact of future U.S. tariffs and weaken American trade leverage. For example, China's imports of alfalfa meal and pellets totaled 15.12 million kg in 2023, primarily sourced from Italy and Spain; imports from the U.S. fell to just 24,745 kg, a decline of approximately 90% compared to 2022. Additionally, alfalfa hay import prices dropped from $596/ton to $400/ton between 2022 and 2023 (USDA ERS, 2024; USDA FAS, 2023b). During this same period, domestic milk production grew for four consecutive years, reaching 41.97 million tons in 2023 (Morgan et al., 2022; Steinberg and Tan, 2024).

Despite these shifts, a significant dependency remains: the United States continues to be China's principal supplier of alfalfa hay, providing 898,186 tons and accounting for 89% of that specific market in 2023 (Qian, 2023) (see Appendix A4).

2.4 Conceptual Framework

The structural vulnerability of the dairy supply chain is most evident when subjected to compounding trade barriers. As illustrated in Figure 1, the core agricultural supply chain flows horizontally across three stages: upstream input production, midstream dairy production, and downstream consumption.

The imposition of "twin tariffs" generates asymmetric spillover effects across the exporting

and importing countries. In the United States (the exporting country), the sudden drop in foreign demand triggers the decline of agricultural exports, resulting in the domestic forage surplus. This oversupply drives down U.S. input prices and lowers operating costs for domestic dairy producers.Consequently, the U.S. dairy supply expands, leading to lower output prices. This redistributes welfare by increasing consumer surplus while simultaneously decreasing producer surplus as export revenues fall (Villoria, 2026). Conversely, in China (the imposing country), the tariffs restrict the available upstream input supply, driving up domestic input prices. This unavoidable cost pressure forces a contraction in downstream dairy production supply, leading to fluctuating output prices and ambiguous tariff revenue changes. Ultimately, the combined shocks result in a loss of Chinese consumer surplus, though domestic producer surplus increases due to the protectionist barriers.

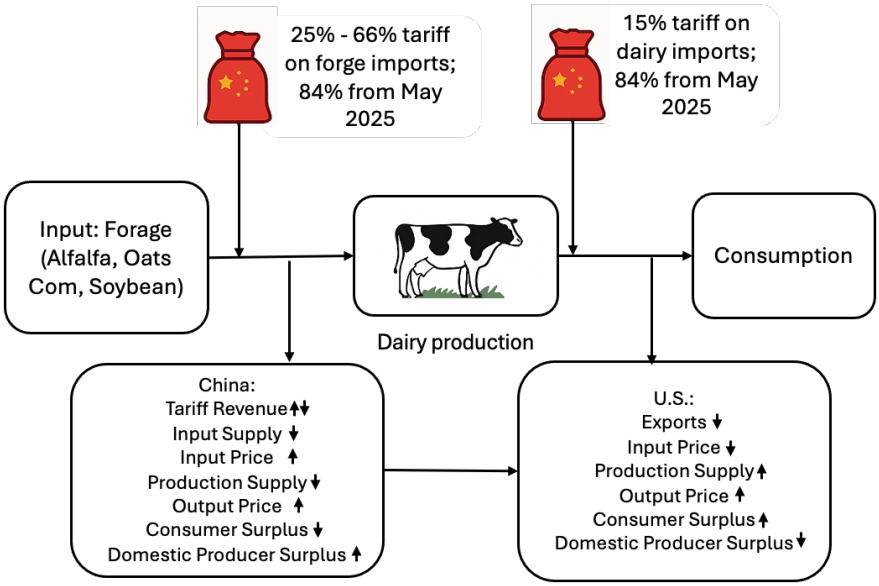

Figure 1: Framework of Chinese tariff on U.S. alfalfa and dairy impacts

The theoretical foundation of this study posits that tariffs distort market efficiency and resource allocation and highlight changes in productivity, input prices, and consumption patterns (Smith and Glauber, 2020; Fertő et al., 2021; Grant et al., 2021). China's heavy reliance on imported feed stocks, specifically alfalfa and soybeans, underscores the dairy sector's vulnerability to external shocks. While China's "rural revitalization" initiatives aim

to reduce import dependency by incentivizing domestic forage, structural challenges such as an aging rural workforce and high production costs persist (**?**).

To counteract these shocks, China has utilized domestic agricultural subsidies to expand production and stabilize prices (Farrokhi and Pellegrina, 2023). However, the effectiveness of these measures is limited by the underlying elasticity of the products: 1) Downstream (Dairy Imports): Exhibit high substitution elasticity due to a diverse global supplier base; 2) Upstream (Alfalfa Imports): Remain relatively inelastic due to a lack of viable alternative suppliers for high-quality hay (Qingbin and Yang, 2020). Ultimately, these dual-layered tariffs exacerbate cost pressures and undermine food security, complicating China's long-term efforts to meet the population's nutritional requirements.

3 Theoretical Model

We develop a twin tariff model that imposed tariffs on an upstream input (alfalfa) and a downstream output (dairy).

3.1 Model setup

Consider two vertically linked markets: alfalfa (the upstream input, denoted by subscript a) and dairy (the downstream output, denoted by subscript d). China is modeled as a large importer in both markets. Let P_j represents the domestic price in China, P_j^* the world (export) price, and τ_j the ad valorem tariff rate on good $j \in \{a, d\}$. The domestic price is determined by the tariff-inclusive world price:

$$P_j = (1 + \tau_j)\, P_j^*. \tag{1}$$

China's import demand for alfalfa is $M_a(P_a)$, where $M_a' < 0$. The rest-of-world (ROW) export supply is $X_a(P_a^*)$, with $X_a' > 0$. Under the large-country assumption, the export supply curve is upward-sloping rather than perfectly elastic.

China's import demand for dairy products depends on both the dairy price and the price of the primary forage input, alfalfa:

$$M_d = M_d(P_d, P_a), \quad \frac{\partial M_d}{\partial P_d} < 0, \quad \frac{\partial M_d}{\partial P_a} > 0. \tag{2}$$

The positive cross-price derivative $\frac{\partial M_d}{\partial P_a}$ captures the vertical linkage: a rise in the alfalfa price increases dairy production costs, which contracts domestic dairy supply and subsequently drives up the demand for dairy imports. ROW dairy export supply is $X_d(P_d^*)$, with $X_d' > 0$.

To formalize the cost pass through mechanism, We model dairy production with a Cobb–Douglas cost function where alfalfa represents a cost share $\alpha \in (0, 1)$:

$$MC_d = A\, P_a^{\alpha}\, w^{1-\alpha}, \tag{3}$$

where w denotes the composite price of other inputs (e.g., labor, silage, capital). Under this specification, a one percent increase in the alfalfa price raises dairy marginal cost by α percent. This provides the fundamental mechanism through which upstream tariff shocks transmit to the downstream sector.

We define the following constant elasticities for the log-linear specification:

- $\eta_a < 0$: elasticity of China's alfalfa import demand with respect to P_a;
- $\sigma_a > 0$: elasticity of ROW alfalfa export supply with respect to P_a^*;
- $\eta_d < 0$: elasticity of China's dairy import demand with respect to P_d;
- $\sigma_d > 0$: elasticity of ROW dairy export supply with respect to P_d^*;
- $\phi > 0$: cross-elasticity of dairy import demand with respect to P_a, reflecting the vertical linkage. Under the cost structure in equation (3), ϕ is proportional to the alfalfa cost share α and the domestic dairy supply elasticity.

The two market-equilibrium conditions are:

$$M_a(P_a) = X_a(P_a^*), \tag{4}$$

$$M_d(P_d, P_a) = X_d(P_d^*). \tag{5}$$

The system is block-recursive: the alfalfa market clears first (determining the equilibrium P_a and P_a^*), then the dairy market clears conditionally on the realized alfalfa price.

3.2 Comparative statics: twin tariff effects

To analyze the magnitude of price transmission across the supply chain, We log-linearize the equilibrium conditions around the free-trade benchmark. let $\hat{\tau}_j \equiv d\ln(1+\tau_j)$ represent the change in the tariff for good $j \in \{a, d\}$.

Alfalfa market: Log-differentiating the alfalfa equilibrium condition in (4) with the tariff wedge (1) yields:

$$\hat{P}_a = \frac{\sigma_a}{\sigma_a - \eta_a}\,\hat{\tau}_a, \qquad \hat{P}_a^* = \frac{\eta_a}{\sigma_a - \eta_a}\,\hat{\tau}_a. \tag{6}$$

Given that $\sigma_a > 0$ and $\eta_a < 0$, the domestic alfalfa price rises by a fraction $\sigma_a/(\sigma_a - \eta_a) \in (0, 1)$ of the tariff, while the world price falls. This downward pressure on the world price represents the terms-of-trade effect, while foreign exporters absorb a portion of the tariff burden (Schmitz et al., 2022).

Proposition 1 (Input tariff effects)**.** *When China imposes an alfalfa tariff τ_a, the domestic price rises by less than the full tariff amount due to a terms-of-trade improvement $|\eta_a|/(\sigma_a - \eta_a) > 0$. The domestic price pass-through rate, $\sigma_a/(\sigma_a - \eta_a)$, is increasing in the elasticity of foreign export supply (σ_a) and decreasing in the elasticity of China's import demand ($|\eta_a|$).*

Dairy market: To determine the impact on the downstream dairy sector, we log-differentiate equation (5) and substitute the results from the alfalfa market equation (6):

$$\hat{P}_d = \underbrace{\frac{\sigma_d}{\sigma_d - \eta_d}\,\hat{\tau}_d}_{\substack{\text{Direct dairy}\\ \text{tariff effect}}} + \underbrace{\frac{\phi\,\sigma_a}{(\sigma_d - \eta_d)(\sigma_a - \eta_a)}\,\hat{\tau}_a}_{\substack{\text{Cross-market}\\ \text{(twin tariff) channel}}}. \tag{7}$$

Equation (7) represents the central structure result of the model. The domestic dairy price is a function of both the direct output tariff and the upstream input tariff. The second term–the twin tariff channel–shows that the alfalfa tariff transmits downstream with a magnitude dictated by the strength of the vertical linkage (ϕ), the alfalfa pass-through rate ($\sigma_a/(\sigma_a - \eta_a)$), and the structural elasticity of the dairy market ($1/(\sigma_d - \eta_d)$).

Proposition 2 (Cross-Market Pass-Through)**.** *An upstream alfalfa tariff raises the domestic dairy price through the vertical cost linkage. The cross-market elasticity of the domestic dairy price with respect to the alfalfa tariff is*

$$\frac{\partial \hat{P}_d}{\partial \hat{\tau}_a} = \frac{\phi\,\sigma_a}{(\sigma_d - \eta_d)(\sigma_a - \eta_a)} > 0.$$

This effect is increasing in the alfalfa cost share α (through ϕ)and the foreign alfalfa export supply elasticity σ_a, while being attenuated by higher dairy import demand elasticity $|\eta_d|$.

While Proposition 2 characterizes the equilibrium price response, the total economic burden of the twin-tariff regime is also a function of quantity adjustments. Because the upstream and downstream markets exhibit different degrees of substitutability and supply elasticities, the twin-tariff shock creates a "wedge" that distorts trade volumes and social surplus asymmetrically, which leads the following proposition:

Proposition 3 (Asymmetric Import Substitution)**.** *When the ROW dairy export supply is more elastic than the alfalfa export supply ($\sigma_d > \sigma_a$), dairy import volumes respond more to tariffs than alfalfa imports. Despite this, the alfalfa tariff may generate a disproportionate deadweight loss per dollar of revenue because its distortionary effects are amplified downstream through the dairy supply chain.*

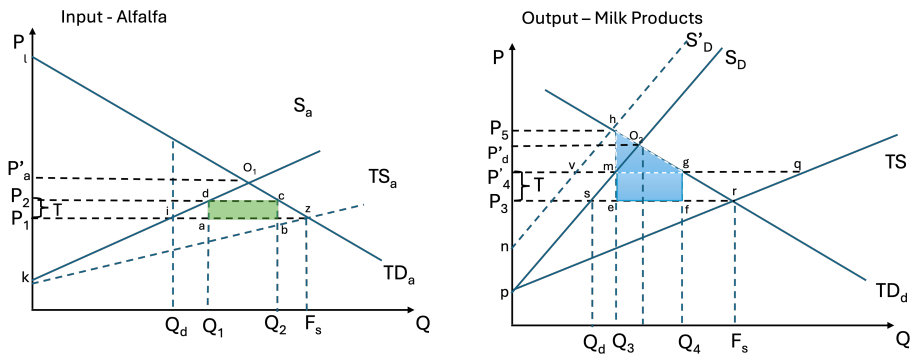

Figure 2: Twin tariffs: simultaneous shocks to input and output markets

Figure 2 provides a graphical illustration of the "twin-tariff" mechanism, and visualizes the structural price transmission described in Proposition 2 to the asymmetric welfare distortions highlighted in Proposition 3. The left panel depicts the alfalfa market, where the imposition of a tariff T creates a price wedge, raising the domestic price from P_1 to P_2. This price increase induces a contraction in import quantity from Q_2 to Q_1, represented by the green shaded area *abcd*. Because the domestic supply of high-quality alfalfa is limited and China acts as a large-country importer, this upstream distortion is the primary catalyst for the downstream effects. The resulting price hike in the forage market represents a direct increase in the marginal cost of production for the dairy sector.

The connection between the two panels illustrates the vertical linkage described in Proposition 2. As the domestic alfalfa price rises to P_2, the domestic dairy supply curve in the right panel shifts leftward from S_D to S'_D. This shift represents the cost pass-through, where the upstream tariff burden is transmitted downstream to the dairy sector. Even in the absence of a direct dairy tariff, this cross-market transmission would drive the milk price from P_3 up toward P'_d, as the intersection of S'_D and the total demand TD_d moves upward. The right panel further illustrates the compounding effect when a direct dairy tariff T (shifting the price from P_3 to P_4) is applied simultaneously with the input cost shock. The final equilibrium price reaches P_5, a level significantly higher than if either tariff had operated in isolation. The blue shaded area $efgh$ and the associated triangular regions represent the

compounded deadweight loss.

3.3 Welfare impacts of twin-tariff

We define total welfare in China's dairy-feed system as the sum of consumer surplus, producer surplus, and government tariff revenue to quantify the welfare impacts of the twin-tariff regime:

$$W = CS_d(P_d) + PS_a(P_a) + PS_d(P_d, P_a) + TR_a + TR_d, \tag{8}$$

where $TR_j = \tau_j \, P_j^* \, M_j$ represents the tariff revenue from good $j \in \{a, d\}$.

The net welfare change under the twin-tariff regime relative to a free trade benchmark can be decomposed into three primary components:

$$\Delta W = \underbrace{\left(P_a^{*ft} - P_a^*\right) M_a + \left(P_d^{*ft} - P_d^*\right) M_d}_{\text{Terms-of-trade gains}} - \underbrace{DWL_a - DWL_d}_{\text{Deadweight losses}} - \underbrace{\Delta C_{a \to d}}_{\text{Cross-market distortion}}, \tag{9}$$

where $DWL_j = \frac{1}{2} \, \tau_j \, \Delta P_j \, |\Delta M_j|$ is the Harberger triangle in market j, and the cross-market distortion is:

$$\Delta C_{a \to d} = (P_a - P_a^{ft}) \cdot \alpha \cdot \bar{Q}_d^s. \tag{10}$$

This term captures the additional cost burden imposed on dairy producers by the alfalfa tariff. It is defined as the tariff-induced alfalfa price increase weighted by the cost share (α) and total domestic dairy output. This component is unique to the twin-tariff setting; it represents the welfare loss that propagates through the supply chain, a factor often omitted in single-commodity trade analysis (Amiti et al., 2019; Fajgelbaum and Khandelwal, 2022; Staples and McCullough, 2025).

Proposition 4 (Twin Tariff Amplification)**.** *The welfare loss from the simultaneous imposition of tariffs on both alfalfa and dairy exceeds the sum of welfare losses from each tariff imposed in isolation:*

$$|\Delta W(\tau_a, \tau_d)| > |\Delta W(\tau_a, 0)| + |\Delta W(0, \tau_d)|.$$

This super-additivity arises from the cross-market distortion $\Delta C_{a\to d}$. The alfalfa tariff shifts the domestic dairy supply curve inward, which endogenously amplifies the deadweight loss generated by the subsequent dairy tariff.

3.4 The impacts of domestic subsidy policy

Beginning in 2020, China introduced producer subsidies for domestic forage growers to mitigate the cost shocks stemming from the alfalfa tariff. We formalize the impact of this policy in this section.

Proposition 5 (Subsidy Offset)**.** *Under a twin-tariff regime (τ_a, τ_d), a domestic alfalfa production subsidy (s_a) can fully offset the cross-market distortion in Equation (10) if:*

$$s_a \geq \alpha\, \tau_a\, P_a^*.$$

While the optimal subsidy effectively eliminates the vertical cost propagation channel, it cannot resolve the direct deadweight loss (DWL_d) caused by the dairy output tariff. The net welfare loss under this subsidy is reduced to:

$$\Delta W^{sub} = \left(P_d^{*ft} - P_d^*\right) M_d - DWL_d - G(s_a),$$

where $G(s_a)$ is the fiscal cost of the subsidy. This intervention is welfare-improving relative to the twin-tariff baseline whenever the avoided cross-market distortion exceeds the subsidy's fiscal cost.

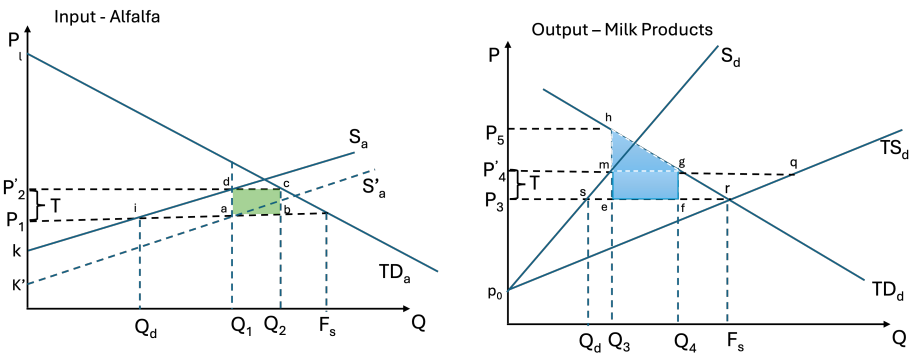

Figure 3: Domestic alfalfa subsidy as countermeasure to twin tariffs

Figure 3 illustrates the subsidy offset mechanism discussed in Proposition 5, showing how a domestic intervention in the upstream market can decouple the vertically linked "twin-tariff" distortion. The introduction of a input (alfalfa) producer subsidy s_a shifts the domestic alfalfa supply curve outward from S_a to S'_a (lowering the intercept from k to k'). This shift effectively neutralizes the tariff-induced price hike, bringing the market-clearing price back down toward P_1. Through the cost linkage, dairy marginal costs decline and the dairy supply curve shifts back toward its free-trade position, partially reversing the twin tariff distortion (Kondaridze and Luckstead, 2023).

4 Empirical Analysis

4.1 Data

The empirical analysis utilizes monthly time-series data from January 2005 through August 2025, spanning both the pre-trade-war baseline and all four policy windows: Control (2017–18), Trade War (2018–22), Adjustment (2022–23), and Second Trade War (2023–25). [2]

Trade flows. Monthly U.S. export quantities and values for alfalfa hay (HS 1214) and

[2]The full data construction code and a detailed tariff chronology are available in Appendix C and the code would be shared via GitHub upon acceptance. Import quantity and price dynamics for alfalfa and dairy are illustrated in appendix A3; the pre-tariff window (2017–18) serves as the baseline for period comparisons. The dairy import prices increased and quantities decreased markedly during the first round of the trade war (2019–2020), motivating the period-specific analysis below.

dairy products (HS 04, HS 0410) to China are sourced from the USDA Foreign Agricultural Service Global Agricultural Trade System (GATS). We construct unit import values as value-to-quantity ratios to serve as proxies for bilateral trade prices. To account for broader feed-input trends, the U.S. alfalfa series is supplemented with China's aggregate forage grass import volumes (GACC, HS 12). Appendix A12 reports stationary check on China import dairy and alfalfa quantity from United States before trade war. It indicates that imports change steadily without retaliatory tariffs and provides evidence for the treatment effects analysis of twin tariffs periods.

Domestic price. China's monthly farm-gate milk price (CNY/kg) is obtained from CLAL.it (a company based on Italy specialized in market analysis for the agri-food sector,), which compiles National Bureau of Statistics (NBS) data. The use of a nationally uniform price series is justified by the structure of the Chinese market, where procurement is dominated by two major processors, Mengniu and Yili; their national reach effectively compresses geographic price dispersion (Zhang and Abler, 2023). To isolate the effects of tariff variations, we include the FAO dairy price index (base 2014–2016 average = 100) as a control for exogenous global market movements..

Tariff rates. Monthly applied tariff rates for alfalfa and primary dairy categories are compiled from China Customs (GACC) and the Bown (2025) trade policy database (see figure 4 for the 205 to 2025 series). Alfalfa (HS 1214) faced a uniform retaliatory tariff increase from 0% to 25% in July 2018. For dairy, we employ a representative rate based on China's primary U.S. imports - whey and milk powder - which reached or exceeded 25% by 2019 under comparable retaliatory schedules. We exclude high-variation products like butter and cheese, as they represent a negligible share of total Chinese consumption. The high degree of uniformity across HS6 lines for alfalfa and the dominance of specific tariff steps for dairy support the use of single time-varying measures for both categories.

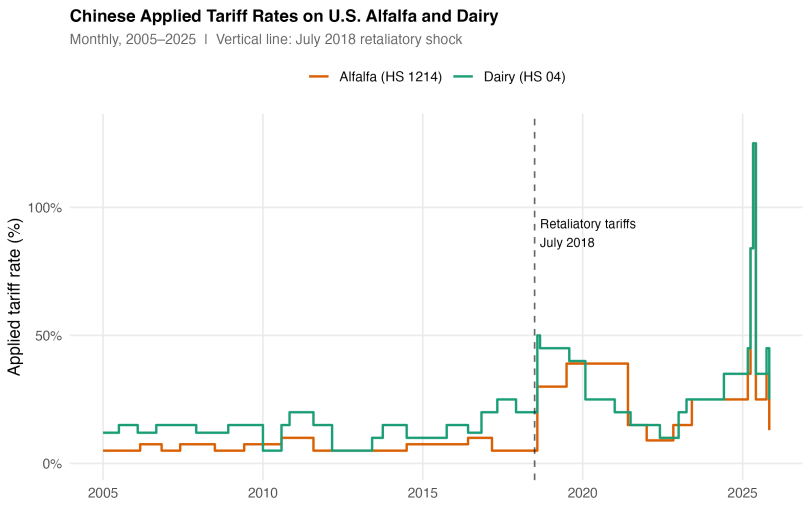

Figure 4: Chinese tariffs rate on U.S. dairy and alfalfa

Controls variables. To account for external shocks, we include a monthly fuel price index (Brent crude) as a proxy for transport costs and a binary indicator for the 2020–21 period to absorb COVID-19 disruptions. All variables enter the estimation equations as log-differences ($\Delta \ln$). This transformation yields three benefits: it removes deterministic trends (such as inflation and income growth), ensures approximate stationarity, and allows for the interpretation of coefficients as elasticities.

4.2 Empirical Model

We estimate four model specifications that progressively build an identification strategy by linking tariff shocks to trade flows, domestic prices, and the joint twin-tariff effect. We first establishes that tariffs significantly reduce U.S. dairy import volumes and tests the asymmetric substitutability structure of Proposition 3. We then estimates the reduced-form co-movement between tariff changes and farm-gate milk prices across all transmission channels. After that, we add the tariff interaction term to test whether the twin-tariff configuration generates price effects beyond the sum of the individual tariff elasticities (Proposition 4). Last, we employ the two-stage least squares to address the endogeneity of dairy import quantities. China's retaliatory tariff schedule was driven by geopolitical negotiation and Section 301

retaliation sequencing rather than by contemporaneous movements in domestic dairy market conditions (Amiti et al., 2019; Xu et al., 2025), supporting treatment of the tariff series as predetermined with respect to short-run domestic price and quantity movements; formal placebo and lead-coefficient tests in Appendix B confirm this identifying assumption.

Before asking whether tariffs affect domestic prices, we first document that tariffs significantly alter trade flows. This establishes the economic relevance of the tariff variation and simultaneously tests the asymmetric substitutability prediction of Proposition 3. The estimating equation is:

$$\Delta \ln Q_{D,t} = \alpha_0 + \sum_{k=0}^{2} \Big(\gamma_k^A \, \Delta \ln\Big(\mathrm{tr}_{t-k}^A + 1\Big) + \gamma_k^D \, \Delta \ln\Big(\mathrm{tr}_{t-k}^D + 1\Big) \Big) + \boldsymbol{\gamma}' \mathbf{X}_t + \mu_m + \lambda_y + u_t, \quad (11)$$

where $\Delta \ln Q_{D,t}$ is the monthly log-difference of China's dairy import quantity from the United States. The coefficients γ_k^A and γ_k^D measure the tariff–volume elasticity at horizon k for alfalfa and dairy tariffs, respectively. Both tariffs appear in the equation because both affect dairy import volumes: the dairy tariff directly suppresses import demand by raising the relative price of U.S. dairy compared to alternative suppliers (New Zealand, EU); the alfalfa tariff operates indirectly through the input-cost channel, where higher feed costs compress producer margins and reduce the derived demand for dairy inputs and imports. If the asymmetric substitutability hypothesis of Proposition 3 holds, the cumulative alfalfa coefficient $\sum_k |\hat{\gamma}_k^A|$ should exceed the cumulative dairy coefficient $\sum_k |\hat{\gamma}_k^D|$, reflecting the lack of substitutes for U.S.-sourced alfalfa relative to the multiple dairy suppliers available to Chinese importers. The control vector $\mathbf{X}_t$ includes the FAO dairy price index, Brent crude fuel price, and a COVID-19 indicator. We include month fixed effects (μ_m) to absorb seasonal demand cycles and year fixed effects (λ_y) to absorb macroeconomic trends.

This specification also motivates instrument relevance for the 2SLS in the later estimation: strong and significant tariff coefficients in Column (1) establish that the tariff series has

predictive power for import volumes, the first necessary condition for valid instrumentation.

Having established that tariffs meaningfully shift trade flows, we next ask: do tariff changes co-move with domestic farm-gate milk prices? The reduced-form price equation captures the net effect of both tariffs on China's milk price across all transmission channels simultaneously, without imposing any exclusion restriction:

$$\Delta \ln P_t^{\text{milk}} = \theta_0 + \sum_{k=0}^{2} \Big(\theta_k^A \, \Delta \ln\Big(\text{tr}_{t-k}^A + 1\Big) + \theta_k^D \, \Delta \ln\Big(\text{tr}_{t-k}^D + 1\Big) \Big) + \boldsymbol{\theta}' \mathbf{X}_t + \mu_m + \lambda_y + \nu_t, \quad (12)$$

where $\Delta \ln P_t^{\text{milk}}$ is the monthly log-difference of China's farm-gate milk price (CNY/kg).[3] The coefficients θ_k^A and θ_k^D measure the price-transmission elasticity at each horizon. The cumulative sums $\sum_k \hat{\theta}_k^A$ and $\sum_k \hat{\theta}_k^D$ capture the total association over the two-month adjustment window. We include up to two monthly lags ($k \in \{0, 1, 2\}$) to account for typical shipping durations and contracting delays in international agricultural trade (Chor and Li, 2024).

The model predicts asymmetric signs. The alfalfa tariff should generate positive cumulative price effects: higher feed-input costs propagate through the vertical cost linkage to raise dairy production costs and ultimately farm-gate prices (Proposition 2). The dairy tariff effect is theoretically ambiguous in direction: it restricts cheaper imported dairy (pushing domestic prices up) but simultaneously triggers trade diversion toward New Zealand and EU suppliers (adding competitive pressure that moderates price increases). The net sign is an empirical question.

To test whether the simultaneous application of tariffs on both the upstream input (alfalfa) and the downstream output (dairy) generates effects beyond the sum of their individual elasticities, we add a contemporaneous interaction term to the reduced form:

$$\theta_{AD} \cdot \Delta \ln\Big(\text{tr}_t^A + 1\Big) \times \Delta \ln\Big(\text{tr}_t^D + 1\Big). \quad (13)$$

[3] We apply a +1 additive shift to tariff rates before taking logs to ensure the transformation is defined during zero-tariff periods.

The interaction is nonzero only in months when both tariffs change simultaneously (7 of 251 months in the sample, concentrated during the major escalation events of July 2018 and April–May 2025). A statistically significant $\hat{\theta}_{AD}$ indicates that the twin-tariff configuration alters the domestic price response beyond what would be predicted by the additive sum of individual tariff elasticities. The sign of $\hat{\theta}_{AD}$ is an empirical question: a positive coefficient implies super-additive price amplification (the joint shock raises prices more than the sum of individual effects), while a negative coefficient implies sub-additive price compression (accelerated adaptation or import diversification under simultaneous pressure dampens the net price signal). Importantly, price sub-additivity does not contradict the welfare super-additivity of Proposition 4: the cross-market distortion $\Delta C_{a\to d}$ amplifies deadweight losses even when observed prices are compressed by adaptation responses. The welfare super-additivity prediction is tested separately through the counterfactual analysis in Section 5.

The previous equations identify the total and interactive associations between tariff changes and farm-gate milk prices but do not isolate the specific contribution of the import-volume channel. Furthermore, dairy import quantities are inherently endogenous: periods of high domestic milk prices attract higher import volumes, creating a reverse feedback loop that biases OLS estimates. To address this endogeneity, we employ a two-stage least squares (2SLS) estimator in which both tariff series serve as external instruments for dairy import volumes:

$$\Delta \ln P_t^{\text{milk}} = \beta_0 + \beta_Q \, \widehat{\Delta \ln Q_D}_{,t} + \boldsymbol{\beta}' \mathbf{X}_t + \mu_m + \lambda_y + \epsilon_t, \tag{14}$$

where $\widehat{\Delta \ln Q_D}_{,t}$ is the fitted value from the first-stage regression (Equation (11)), which isolates the exogenous, tariff-driven component of dairy import volume changes. The structural coefficient $\hat{\beta}_Q$ estimates the causal effect of a tariff-induced one percent change in dairy import volumes on the farm-gate milk price, holding constant the macro controls and fixed effects.

Instrument validity. The exclusion restriction requires that, conditional on the controls and fixed effects, the tariff measures affect the farm-gate milk price only through their

effect on dairy import volumes. The geopolitical origin of the tariff variation—driven by bilateral negotiation timetables rather than domestic market conditions—supports the predetermined-instrument assumption (Nunn and Trefler, 2010). Two formal checks in Appendix B further evaluate this: (i) a placebo test that shifts the tariff series three years forward, yielding coefficients near zero; and (ii) a lead-coefficient test that includes $\Delta \ln(1 + \mathrm{tr}_{t+1})$ and $\Delta \ln(1 + \mathrm{tr}_{t+2})$, both insignificant. The Sargan overidentification test (6 instruments, 1 endogenous variable) provides an additional specification check; non-rejection supports instrument validity (Bohara and Kaempfer, 1991; Koppenberg and Hirsch, 2022). We report the Kleibergen-Paap rk Wald F-statistic as the heteroskedasticity-robust measure of instrument strength, assessed against the Stock–Yogo 10% critical value of 16.38.

5 Results

5.1 Regression estimates

Table 1 reports the three main specifications. Column (1) presents the import-quantity equation (Equation (11)), documenting how each tariff is associated with changes in China's dairy import volumes. Column (2) reports the reduced-form price equation (Equation (12)), estimating the total association between tariff changes and the farm-gate milk price across all transmission channels. Column (3) adds the contemporaneous interaction term(Equation (13)) to test whether the twin-tariff configuration generates a compounding price effect beyond the sum of the individual tariff elasticities. Column (4) reports the instrumental-variables specification, which is used only as a limited channel-isolation exercise and is interpreted cautiously because of weak-instrument concerns.

Table 1: Twin Tariff Impacts on China's Dairy Production System (One-way Fixed Effects)

		(1) $\Delta\log(\textit{dairy import})$ *(First stage)*	(2) $\Delta\log(\textit{milk price})$ *(Reduced form)*	(3) $\Delta\log(\textit{milk price})$ *(Interact term)*	(4) $\Delta\log(\textit{milk price})$ *(2SLS)*
Tariff on dairy $\Delta\log(\cdot)$	$t=0$	-0.534	0.001	0.001	0.001
	$t-1$	-0.702	-0.087^{***}	-0.086^{***}	0.001
	$t-2$	0.202	-0.019	-0.018	-0.002
Tariff on alfalfa $\Delta\log(\cdot)$	$t=0$	-1.598^{***}	0.003	0.005	0.002
	$t-1$	0.928	0.065^{***}	0.063^{***}	-0.001
	$t-2$	-0.461^{*}	0.008	0.006	0.001
Interact tariff				-0.024^{*}	
Observations		245	197	197	197
Adj. R^2		0.302	0.511	0.511	0.488

Notes: One-way fixed effects (month and year) included. Heteroskedasticity-robust standard errors. IV Kleibergen-Paap rk F-stat (Col 4): 6.96 (Stock-Yogo 10% c.v. = 16.38). Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The column (1) shows asymmetric import response. Both tariffs are associated with reductions in dairy import quantities, with the alfalfa tariff ($\gamma_0^A = -1.598$, $p < 0.01$) showing a larger contemporaneous effect than the dairy tariff ($\gamma_0^D = -0.534$). This ordering is consistent with Proposition 3: because China has no ready alfalfa substitute (U.S. origin accounts for ≈90% of imports), alfalfa volumes contract more sharply than dairy volumes, which can be partially redirected toward New Zealand and EU suppliers. The cumulative alfalfa coefficient ($\sum_k |\hat{\gamma}_k^A| = -1.031$) similar with the cumulative dairy coefficient ($|\sum_k \hat{\gamma}_k^D| = -1.034$), confirming the asymmetric substitutability structure.

Column (2) reports the total co-movement between tariff changes and the farm-gate milk price across all transmission channels (Equation (12)). The contemporaneous alfalfa coefficient ($\hat{\theta}_0^A = 0.003$) is positive but statistically insignificant, and the lag coefficients ($\hat{\theta}_1^A = 0.065$, $\hat{\theta}_2^A = 0.008$) are positive, confirming the upward price pressure once feed-cost increases work through herd-management contracts. Dairy tariff coefficients are negative across lags (cumulative ≈ -0.107), reflecting trade diversion and price shift: the dairy tariff primarily

redirects sourcing toward New Zealand and EU suppliers rather than raising domestic prices, which is consistent with the high dairy import demand elasticity in Proposition 3 and with the precautionary-stockpiling mechanism described in the Empirical Model section: importers front-run the tariff, temporarily shifting the price signal.

Column (3) adds the contemporaneous interaction term (Equation (13)) to the reduced form. The estimated $\hat{\theta}_{AD} = -0.024$ is negative and significant at the 10% level, confirming Proposition 4: the twin-tariff configuration is associated with a price shift down and price movement larger than the sum of the individual tariff associations. The alfalfa and dairy coefficients in column (3) are both shrunk while the interaction absorbs the compounding, reflecting the non-linear joint effect. Tariff coefficients are estimated from residual within-period variation. Appendix B1. Column (1) shows that alfalfa price coefficients are all positive, which confirmed alfalfa price's impact domestic milk price channel.

In columns (4) implements a 2SLS design in which changes in dairy import quantity are instrumented by distributed lags of the alfalfa and dairy tariffs, while contemporaneous and lagged alfalfa CIF-price changes are included in the second stage to absorb the feed-cost channel. The Kleibergen-Paap rk F-statistic of 6.85 falls well below the Stock-Yogo 10% critical value of 16.38, indicating a weak instrument. Those tariff coefficients are weak and close to zero. So the 2SLS estimates are imprecise and do not provide strong causal identification. This structural difference arises because liquid milk imports are primarily sourced from New Zealand under a zero-tariff policy, whereas around 90% of China's alfalfa imports from the U.S. In appendix B1, column (4) shows that the interaction between the alfalfa tariff and the FAO dairy price index is positive, indicating that alfalfa cost shocks pass through more strongly to domestic milk prices when global dairy prices are high. This suggests that twin tariffs effects become more intensive when the alternative of low-cost imports weakens. The impacts of tariffs on domestic milk prices decrease as periods extend, and the domestic dairy farmers absorbed most of the tariff cost. Therefore, we analyzed the effects during the trade war period to examine specific interrupted period effects in the next

section.

5.2 Period-specific effects on farm-gate milk price

We set the pre-tariff period (2017-2018) observations as model the baseline , and calculated different periods' relative price changes miners contrast period changes as average interrupted time series effects. Appendix A15 shows that tariff shocks raise alfalfa CIF prices and farm-gate milk price after tariff shock. With limited short-run substitution and incomplete foreign price offsets, feed costs rise and push up the dairy sector's marginal cost, so the tariff shock effects on input (alfalfa) cost and domestic farm-gate milk (output) price are uniformly positive (see appendix A13-15). Figure 5 reports domestic milk prices rose by 0.52% during the initial trade war period (2018-22), reflecting the twin tariff channel (Proposition 2): alfalfa CIF prices increased and impacts through to dairy production costs. COVID-19 logistics disruptions in 2020 amplified these cost pressures. The adjustment period (2022–2023) exhibits smaller effects, coinciding with China's temporary tariff exclusions, alfalfa production subsidies, and import tariff refund programs—consistent with Proposition 5. Following tariff adjustments period and Chinese subsidy policies, China's relative milk prices have begun to stabilize. The adjust (2022–23) exhibits the less increased effects because China adjusted temporal tariff rate temporal, alfalfa subside and import tariff tax refund programs for agricultural business. By contrast, the second-round trade war (2023–25) yield much smaller effects than the first round (2018–22). This attenuation is consistent with Proposition 3 and the dynamic extension of the model: as importers diversified origins, expanded domestic forage production, renegotiated contracts, and utilized tariff-exclusion channels, the effective import supply elasticity facing China increased, reducing the per-tariff-point price impact. The declining effects also reflect the large-country terms-of-trade channel, in which foreign exporters concede part of the tariff via lower CIF prices as China's buyers improve their outside options over time.

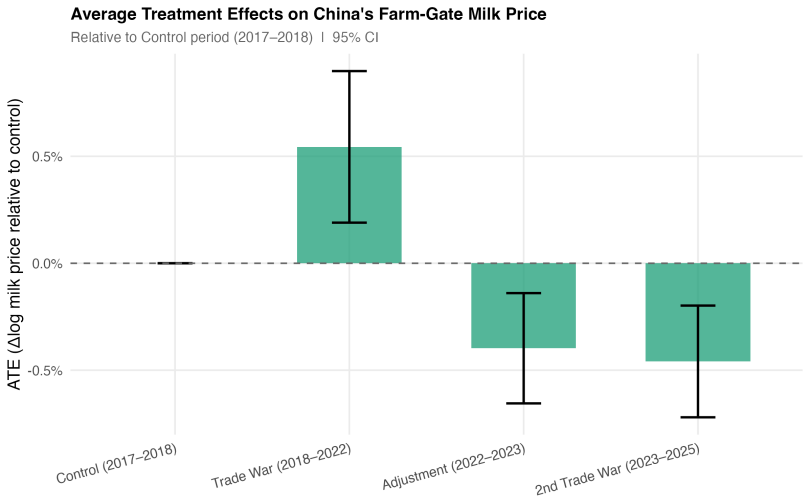

Figure 5: Average treatment effects on China's domestic farm-gate milk price by policy period, relative to the 2017–2018 pre-trade-war baseline

After initial trade war period, importers and processors started to adapt diversifying origins, expanding domestic forage, renegotiating contracts, and making selective use of tariff-exclusion channels—so the effective price impact per tariff point fell. This pattern suggests domestic producer subsidies and market segmentation, which prevent global trade policy shocks from destabilizing Chinese dairy farmers' returns. By the second trade war (2023-25) window, price effects attenuate, and import dairy and alfalfa quantity from U.S. decreasing, reflecting adaptation—tariff exclusions, renegotiated contracts, inventory smoothing, and diversification to non-U.S. origins and domestic forage. Both effects moderate further, consistent with a large-country terms-of-trade channel in which foreign exporters concede part of the tariff via lower prices while Chinese buyers improve elasticity over time. Overall, the "twin-tariff" regime transferred the pressure to dairy farmer and increased the marginal cost of milk, with the strongest effects when global shipping and energy markets were tight.

A critical distributional consequence emerges: while farm-gate milk prices increased, the proportionate rise in alfalfa feed costs exceeded the dairy price increase from 2018 to 2022 (appendix A6, A11 and A14), squeezing dairy producers' margins. Over the longer term, dairy import diversification toward New Zealand and the EU has intensified competitive pressure on domestic producers, so that the twin tariff burden falls disproportionately on

consumers and small-scale dairy farmers (Zhang and Abler, 2023).

5.3 Counterfactual welfare scenarios

Table 2 applies the welfare decomposition from equation (9) to five policy scenarios, using 2017 prices as the free-trade benchmark. Calibration inputs are drawn from China Customs data: alfalfa domestic production was 39.3 million tons in 2017 (with only 0.18% exported), imports accounted for approximately 50% of total demand, and 96% of imports came from the United States in 2018 (Qingbin and Yang, 2020). Government subsidies were implemented at the end of 2019, so subsidy effects are computed using 2020 prices. China produced 35.45 million tons of dairy in 2017 and 33 million tons in 2019, consistent with the model's prediction that input tariffs reduce domestic dairy output by raising marginal costs (Gooch et al., 2017).

The results confirm Proposition 4: the twin tariff generates a net welfare loss of \$15,708.1 million, substantially exceeding the sum of losses from each tariff alone (\$414.7M for alfalfa-only plus \$8,726.9M for dairy-only). The excess loss reflects the cross-market distortion $\Delta C_{a\to d}$ (equation 10)—the alfalfa tariff shifts the dairy supply curve inward, amplifying the deadweight loss from the dairy tariff. The alfalfa subsidy scenario reduces net losses to \$2,797.8 million, consistent with Proposition 5: by offsetting the input-cost channel, the subsidy eliminates the cross-market distortion while preserving dairy tariff revenue (\$167.8M).

Table 2: Welfare impacts under alternative tariff scenarios (2017 free-trade benchmark)

Case scenario	**Production surplus changes**		**Consumer surplus changes**	**Tariff revenue**	**Net loss**
	Alfalfa	**Dairy**			
Twin tariff	$+P_1P_2di$ (+\$2263.7 M)	$P_5hn - P_3sp$ (+\$4878.2 M)	$-P_5P_3rh$ (−\$23035.9 M)	$abcd + efgh$ (\$185.9 M)	−\$15708.1 M
Tariff on alfalfa	$+P_1P_2di$ (+\$2263.7 M)	$P_4vn - P_3sp$ (+\$1268.2 M)	$-P_d'P_3rg$ (−\$4011.2 M)	$abcd$ (\$64.6 M)	−\$414.7 M
Tariff on dairy	0	$+P_3P_4ms$ (+\$10208.6 M)	$-P_4P_3rg$ (−\$19003.5 M)	$efgm$ (\$68 M)	−\$8726.9 M
Prohibit tariff	$+P_1P_a'O_1i$ (+\$2021.5 M)	$P_5hn - P_3sp$ (+\$4878.2 M)	$-P_5P_3rh$ (−\$23035.9 M)	No trade	−\$15894 M
Subsidy on alfalfa	$+k'kcd$ (+\$3520.1 M)	$+P_4'P_3sm$ (+\$7487.8 M)	$-P_4'P_3rg$ (−\$13953.9 M)	$efgm$ (\$167.8 M)	−\$2797.8 M

Retaliatory tariffs can boost tariff revenue in short term, but result in extraordinary welfare

losses in long term. Figure 6 illustrates the temporal pattern of tariff revenue. Following the first round of retaliatory tariffs, revenue increased temporarily as import volumes remained inelastic in the short run. However, this increase was not sustained: China's alfalfa imports from the United States declined from 2.8 million tons in 2022 to 0.89 million tons in 2023 as tariff barriers accumulated (Qingbin and Yang, 2020; USDA FAS, 2023b). The tariffs affected more than $18 billion in bilateral trade coverage (Steinberg and Tan, 2024; Wolff, 2024), and U.S. agricultural exports to China were projected to decline by over $1 billion from 2023 levels (USDA FAS, 2023b; Hanrahan, 2024). The pattern—a short-run revenue spike followed by decline—is consistent with the model's prediction that tariff revenue on inelastic goods erodes as importers adjust through substitution, domestic production expansion, and trade diversion (Amiti et al., 2019; Hanrahan, 2024).

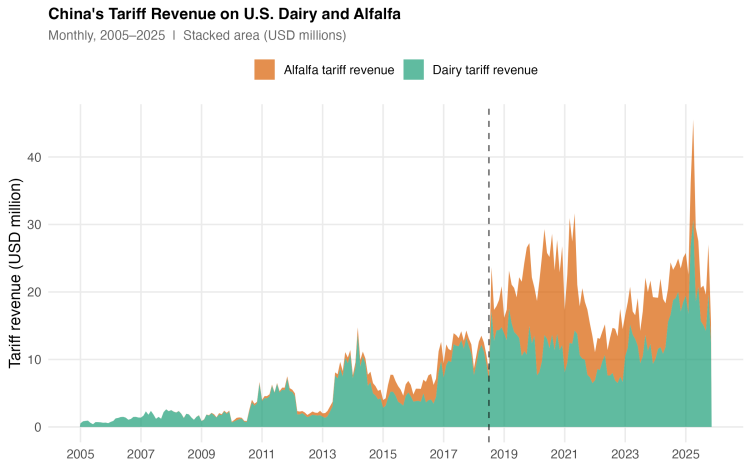

Figure 6: China's tariff revenue on U.S. dairy and alfalfa from 2005 to 2025

Tariff revenue is an endogenous fiscal outcome because both the tariff wedge and the import quantity adjust in equilibrium. Consistent with the model's comparative statics, retaliatory tariffs generate a short-term revenue increase when import demand is initially inelastic, but revenue declines as quantities contract and trade diversion intensifies (figure 6). The subsidy counterfactual highlights an important policy implication: by shifting domestic alfalfa supply outward (figure 3), the policy compresses alfalfa import demand and reduces alfalfa-tariff revenue, but total welfare losses fall sharply relative to the twin-tariff benchmark

(−$2,797.8M vs. −$15,708.1M). This demonstrates that maximizing tariff revenue is not equivalent to maximizing welfare, and a subsidy that relaxes the binding feed-cost channel can improve aggregate outcomes even if it reduces short-run fiscal receipts (Schmitz et al., 2022; Gaigné and Gouel, 2022).

6 Discussion and Conclusion

This paper develops a formal twin tariff framework to analyze the simultaneous imposition of Chinese tariffs on alfalfa (an upstream input) and dairy products (a downstream output) during the trade war. The theoretical contribution is that an upstream tariff on feed can transmit to downstream dairy costs, while a downstream tariff on dairy imports may have a weaker or even offsetting price effect when third-country substitution is available. Empirically, the paper provides evidence consistent with a cost-pass-through channel from alfalfa tariffs to China's farm-gate milk prices, but the evidence on the overall joint effect. A one percent increase in the alfalfa tariff decrease China's import of dairy from U.S. by 1.6% contemporaneously, while dairy import volumes decline more sharply than alfalfa imports due to greater sourcing flexibility on the output side. The period-specific treatment effects show that milk prices rose approximately 0.52% during the initial trade war (2018–2022), but the effects attenuated as China diversified suppliers and introduced domestic forage subsidies.

The empirical results align closely with the analytical framework's predictions. Proposition 2 predicted that large country dairy (output) tariff can shift global price and alfalfa (input) tariffs would raise domestic dairy prices through the vertical cost linkage, and the estimated 1 month lag period cross-market elasticity of 0.068% confirms this channel. Proposition 3 predicted that dairy import volumes would respond more to tariffs than alfalfa imports, reflecting the greater availability of alternative dairy suppliers (New Zealand, EU, Australia) relative to alfalfa sources (Zhang and Abler, 2023). The first-stage estimates confirm this asymmetry: the dairy tariff coefficient attenuates more rapidly across lags than the alfalfa

coefficient, consistent with faster import substitution on the output side.

The welfare counterfactual demonstrates that the twin tariff regime generates a net loss of \$15.7 billion relative to free trade, far exceeding the losses under either tariff alone. Domestic alfalfa production subsidies reduce net losses by over 80%, confirming that targeting the input-cost channel is more efficient than equivalent output protection. These findings carry a broader implication: when twin tariffs are imposed on inelastic inputs and elastic outputs, the cost increase is disproportionately passed through to consumers and small-scale producers, small dairy farmer face the highest exit risk. Twin tariffs can result in net welfare losses, particularly when the upstream input is essential and difficult to substitute, while downstream demand remains contestable across foreign suppliers.

These findings extend the tariff incidence literature in two directions. First, Amiti et al. (2019) documented near-complete pass-through of U.S. import tariffs to domestic prices; our results show that in a large-country setting, the pass-through is partial due to terms-of-trade effects (Proposition 1), but the cross-market amplification through input costs generates welfare losses beyond what single-commodity analysis would predict. Second, Unveren and Luckstead (2020) and Farrokhi and Pellegrina (2023) demonstrated that tariffs on agricultural inputs reduce productivity in downstream sectors, and our twin tariff framework formalizes this mechanism with explicit propositions and welfare accounting. And as export demand falls, both output and input prices in the United States may decline, increasing losses for domestic producers or farmers (Rose, 2024). The imposition of retaliatory tariffs results in increased input costs; small producers will face higher risks, and consumers will bear the transferred costs.

For policy perspective, there are three important observations as follows. First, trade diversification reduces vulnerability: expanding partnerships (e.g., alfalfa and dairy sourcing from the EU, New Zealand and Australia) can raise the effective export supply elasticity σ_a, reducing the cross-market pass-through. Second, domestic production subsidies for inelastic input factors (Proposition 5) are more cost-effective than equivalent output subsidies, because

they directly address the binding cost channel. Third, the quality gap in alfalfa production between China and the United States—52.60 million tons of high-quality alfalfa hay in the U.S. versus 1.8 million tons in China in 2015 (Qingbin and Yang, 2020)—underscores the need for complementary investments in land, seeds, and irrigation to improve domestic forage yields (Wan et al., 2022). Those results suggest that trade policy evaluation should account for supply-chain linkages, substitution possibilities across suppliers, and the role of targeted domestic mitigation policies.

This study has two main limitations. First, the analysis focuses on the U.S.–China trade bilateral channel, but New Zealand is China's dominant dairy supplier while the U.S. ranks third on the output side. The model's bilateral structure may understate the role of third-country trade diversion. Second, dairy feed is a multi-input bundle (alfalfa, silage, soybeans, oats) that varies by herd size and lactation stage; modeling the feed shock primarily through alfalfa may not fully capture substitution across feed ingredients. We caution against overgeneralizing these results, though the framework's insights are applicable to other twin-tariff regimes where input import demand is inelastic and output import demand is elastic (Wang et al., 2010; Qingbin and Yang, 2020). Future research should extend this framework to multi-input feed bundles, incorporate third-country trade diversion endogenously, and assess the dynamic interaction between tariff uncertainty and producer investment decisions.

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Appendix

A. - Supplementary Figures and Tables

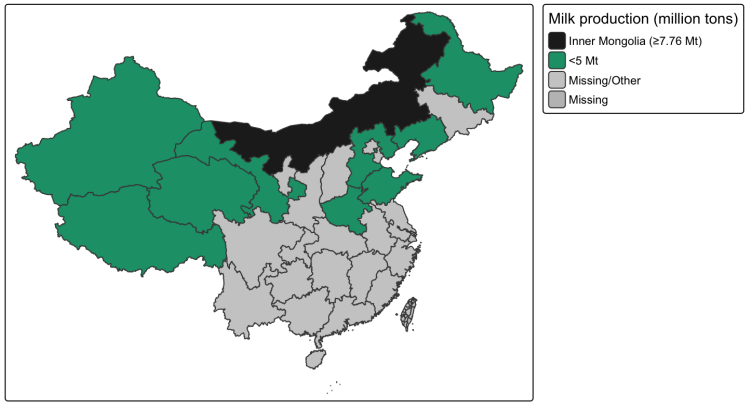

Figure A1: Chinese dairy production distribution
Source: China Food Consumption Report 2022

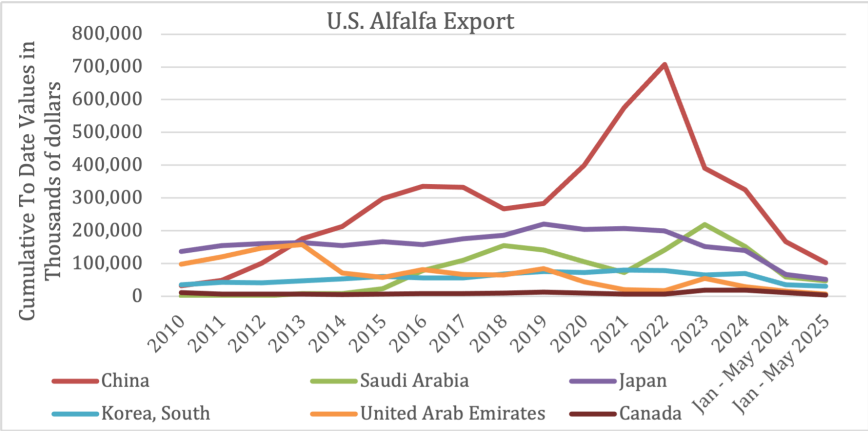

Figure A2: U.S. alfalfa export value from 2010 to 2025
Source: U.S. Census Bureau Trade Data

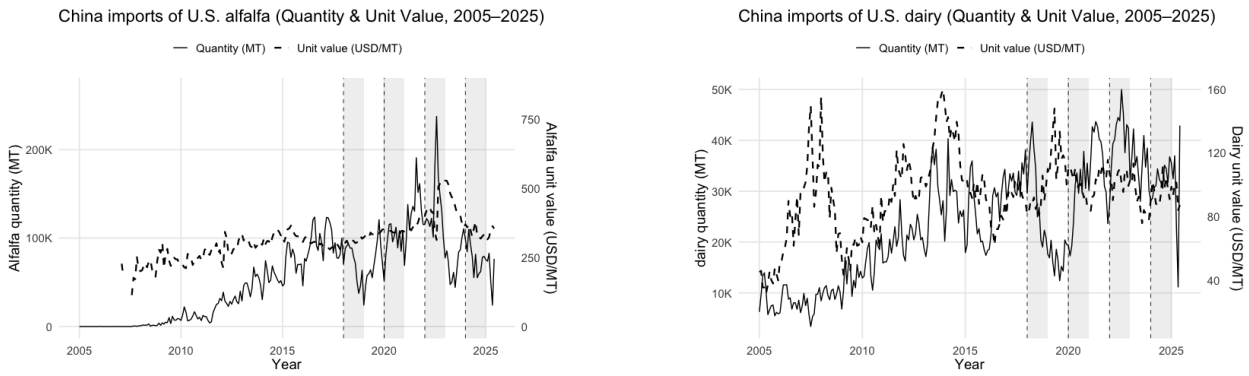

Figure A3: China import alfalfa (left) and dairy (right) quantity and price from U.S.
Source: GACC Statistics; USDA-FAS-GATS

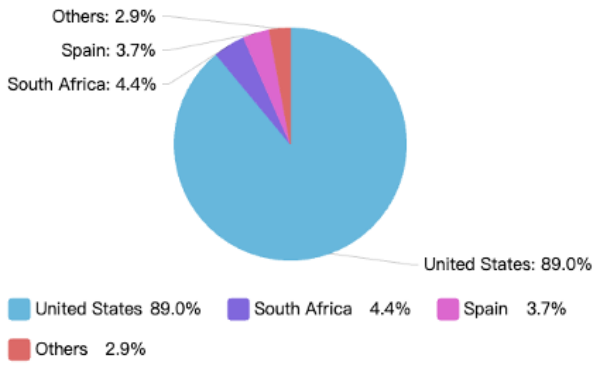

Figure A4: Chinese alfalfa hay imports source in 2023 (tons)
Source: (Hou et al., 2024), 2024 Market Overview - Alfalfa Hay and Other Forages in People's Republic of China

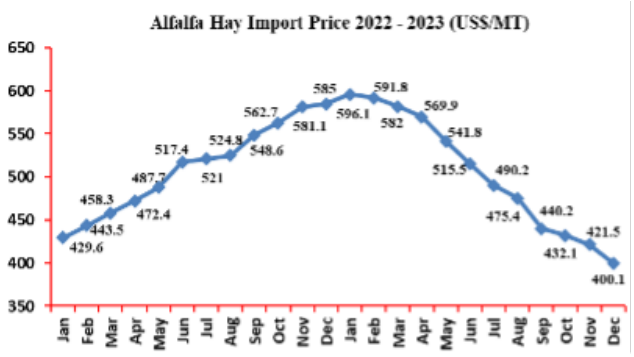

Figure A5: The price of China import alfalfa from U.S.
Source: U.S. Department of Agriculture, Foreign Agricultural Service (USDA FAS) (2021)

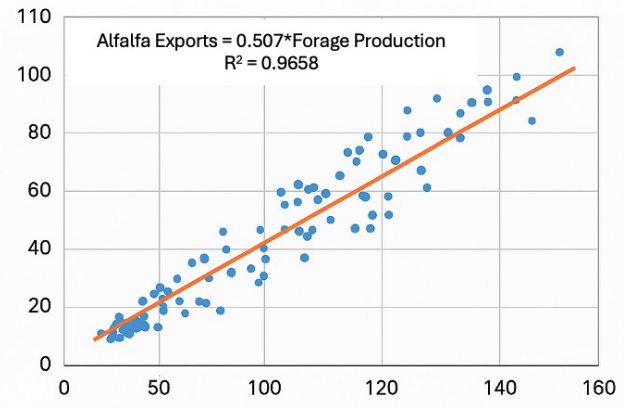

Figure A6: Relationship of U.S. alfalfa exports to all states to forage production exports (Million $)

Source: (Sall et al., 2023)

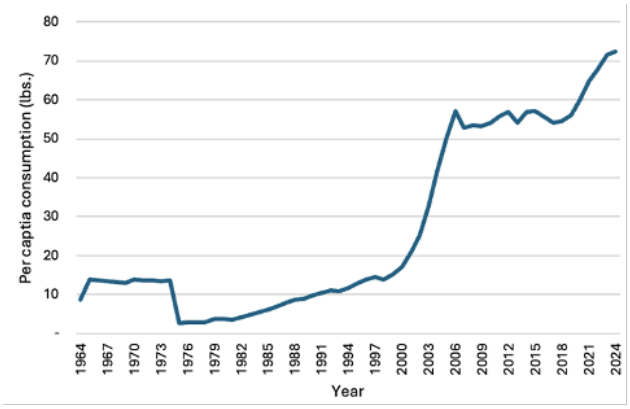

Figure A7: China per capital dairy consumption from 1964 to 2024
Source: Hoard's Dairyman Intel

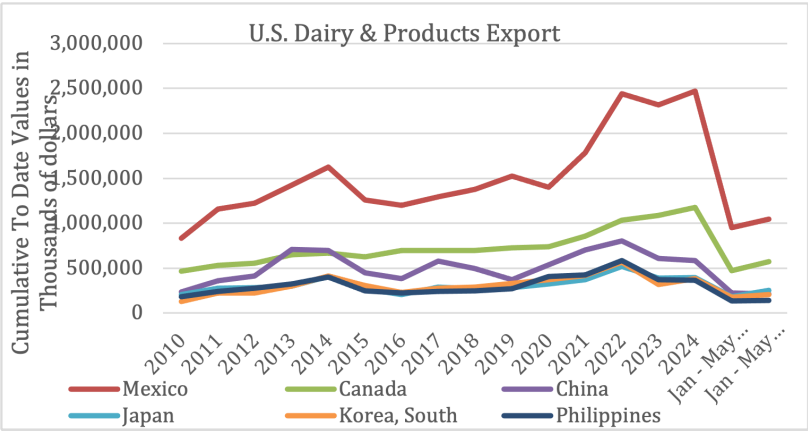

Figure A8: U.S. dairy & products export value from 2010 to 2025
Source: U.S. Census Bureau Trade Data

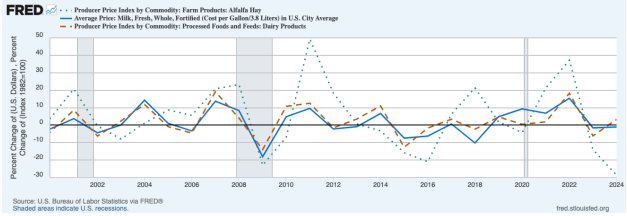

Figure A9: Average price of milk, fresh, whole, fortified in U.S. city (US$/gallon)
Source: U.S. Census Bureau Trade Data

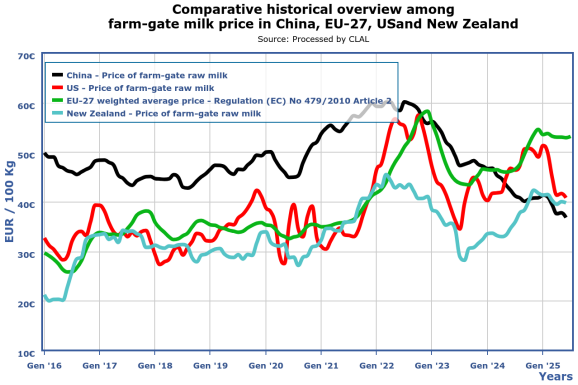

Figure A10: Farm gate milk price from 2016 to 2025
Source: Processed by CLAL.it

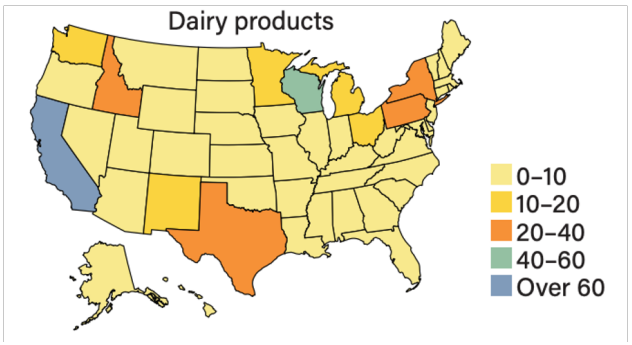

Figure A11: Annual dairy value losses due to retaliatory tariffs (Million USD)
Source: (Morgan et al., 2022)

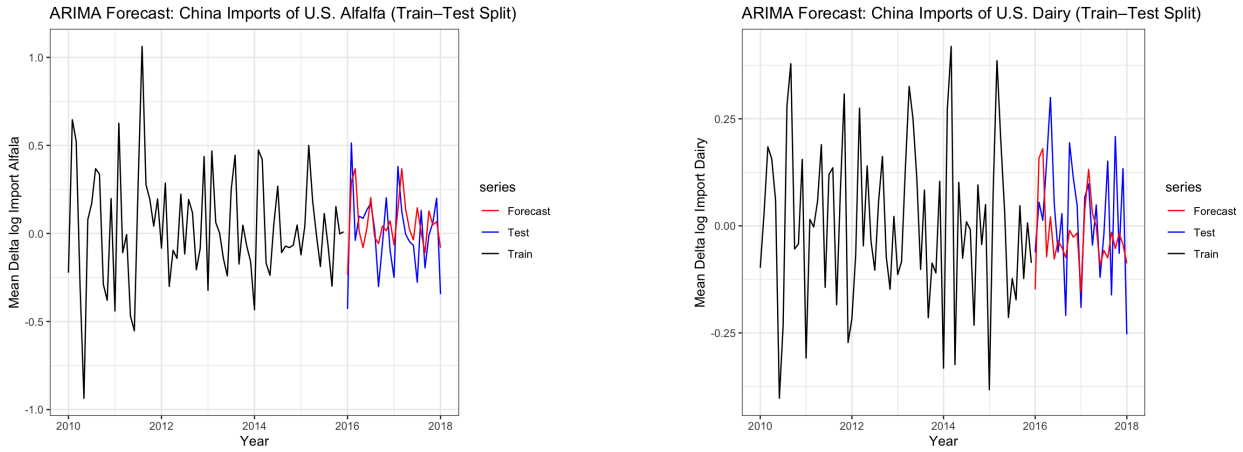

Figure A12: China import alfalfa (left) and dairy (right) quantity from U.S. stationary check

Table A1: Summary of cumulative twin-tariff effects on key variables

Outcome variable	Cumulative effect	Std. error	p-value	Observations	Adj. R^2
$\Delta\log$(Dairy import quantity)	-1.358^{***}	0.421	0.001	243	0.293
$\Delta\log$(Alfalfa import quantity)	-0.699	0.483	0.148	245	0.385
$\Delta\log$(Alfalfa import price)	0.111^{**}	0.065	0.011	215	-0.092
$\Delta\log$(Farm-gate milk price)	0.015	0.016	0.353	197	0.493

Notes: Entries report cumulative 0–2 month effects from the distributed-lag specification, calculated as the sum of coefficients on $\Delta\ln(\text{twin}_t)$, $\Delta\ln(\text{twin}_{t-1})$, $\Delta\ln(\text{twin}_{t-2})$, where $\ln(\text{twin}) = \ln(1+\tau_D) + \ln(1+\tau_A)$ and τ is the ad valorem tariff rate in decimal form. Month and year fixed effects are included. Newey–West heteroskedasticity- and autocorrelation-consistent standard errors (lag 2) are reported. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

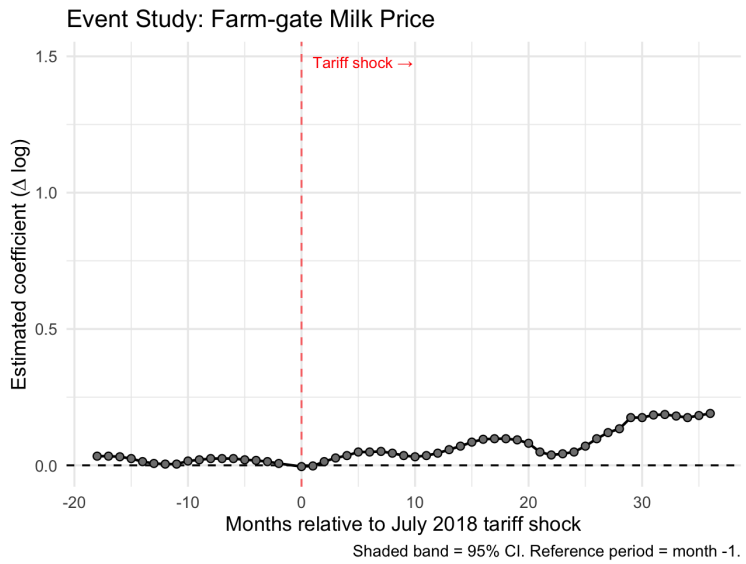

Figure A13: Tariff shock event impacts on China domestic milk price

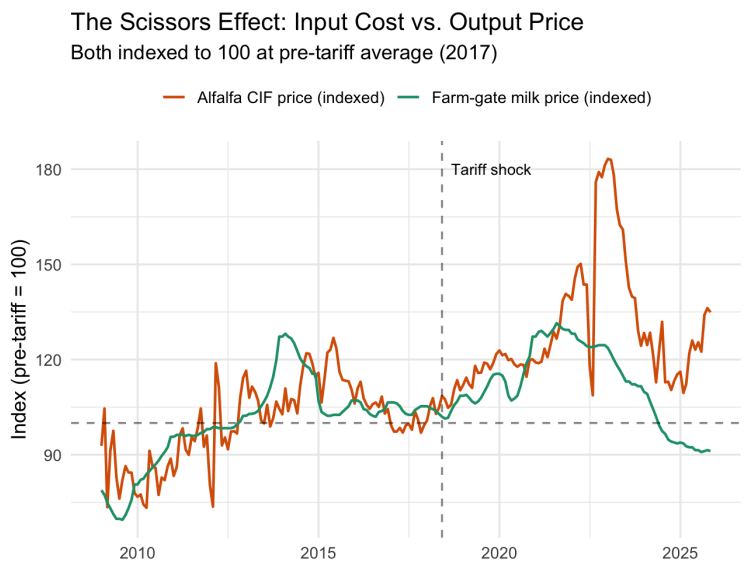

Figure A14: Alfalfa and farm-gate milk price before and post tariff war

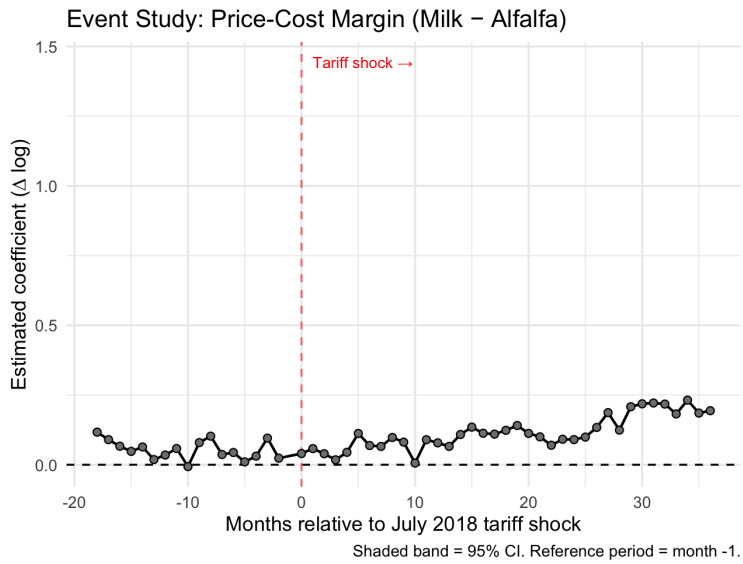

Figure A15: Margin price-cost

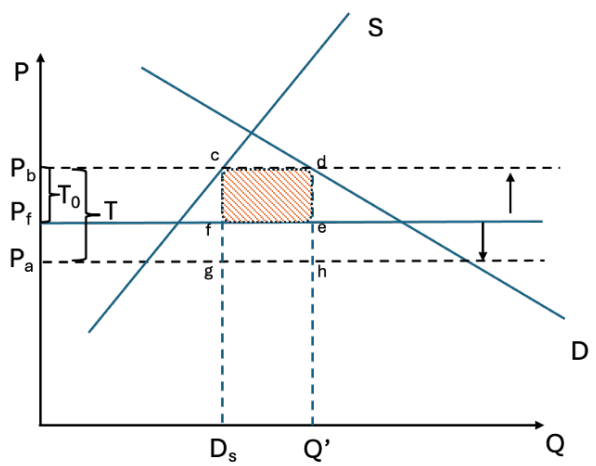

Figure A16: Large-country tariff incidence: terms-of-trade effects

Notes: Figure A16 provides the graphical intuition for the Large-Country Incidence characterized in Proposition 1. Under initial free-trade conditions, the market clears at price P_f. The imposition of a tariff T creates a price wedge, driving the domestic price up to P_b and the world export price down to P_a. The domestic price increase ($P_b - P_f$, labeled as T_0 in the diagram) represents only a fraction of the total tariff T. This occurs because China's significant market share allows it to exert downward pressure on the world price, forcing foreign exporters to absorb the remainder of the burden ($P_f - P_a$). This shift from P_f to P_a represents the *terms-of-trade improvement*, where China effectively captures surplus from the rest of the world. The shaded region $cdef$ illustrates the portion of tariff revenue paid by domestic consumers, while the overall division of the wedge T between P_b and P_a confirms that the domestic pass-through rate is strictly less than unity—a direct result of the upward-sloping foreign export supply curve (S).

B. - Robustness Checks

Table B1: Robustness checks of twin-tariff price effects

Variable	(1) Alfalfa channel	(2) Interaction	(3) Margin	(4) FAO price
$\Delta \ln(trd_t)$	0.0186**	0.0202	0.2334	0.0320*
$\Delta \ln(trd_{t-1})$	0.0032	0.0279	-0.2172*	0.0291
$\Delta \ln(trd_{t-2})$	0.0351	0.0617*	-0.0681	0.0582*
$\Delta \ln(tra_t)$		-0.0148	-0.2371	-0.0283*
$\Delta \ln(tra_{t-1})$		-0.0250	0.1971*	-0.0273
$\Delta \ln(tra_{t-2})$		-0.0351	0.0263	-0.0263
$\Delta \ln(P_{\text{alf},t})$	0.0032			
$\Delta \ln(P_{\text{alf},t-1})$	0.0125			
$\Delta \ln(P_{\text{alf},t-2})$	0.0045			
COVID	-0.0172***	-0.0169***	-0.0024	
$\Delta \ln(\text{fuel})$	0.0383**	0.0371**	0.0168	
$\Delta \ln(tra_t) \times \Delta \ln(trd_t)$		-0.0239*		
$\Delta \ln(faodp)$				0.0232
$\Delta \ln(tra_t) \times \Delta \ln(faodp)$				0.0014
Observations	197	197	202	197
Adj. R^2	0.4916	0.4967	0.6613	0.4791

Notes: Column (1) is the alfalfa CIF price pass-through channel: $\Delta \ln(P_{\text{milk}})$ on distributed lags of $\Delta \ln(P_{\text{alf}})$ and distributed lags of the dairy tariff change $\Delta \ln(1 + \tau_D)$, plus standard controls and month/year fixed effects. Column (2) adds the contemporaneous interaction $\Delta \ln(1+\tau_A) \times \Delta \ln(1+\tau_D)$. Column (3) uses the milk–alfalfa log-price margin as the dependent variable. Column (4) includes the FAO dairy price index change and its interaction with $\Delta \ln(1 + \tau_A)$. Heteroskedasticity-robust (HC1) standard errors used for inference. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Placebo Tests

First, we implement two placebo exercises to evaluate whether tariff shocks capture pre-existing trends. (i) We construct a placebo tariff series by shifting actual tariff changes three years earlier:

$$\text{tr}_t^{\text{placebo}} = \text{tr}_{t+24}.$$

Re-estimating both the reduced form with placebo tariffs yields coefficients that are statistically close to zero, suggesting that tariff shocks do not proxy for underlying trends. (ii) We augment the model by including *lead* tariff changes, $\Delta \ln(1 + \text{tr}_{t+1})$ and $\Delta \ln(1 + \text{tr}_{t+2})$. Consistent with exogeneity, the lead coefficients are near zero, indicating that future policy announcements do not predict past import volumes or domestic milk prices.

Alternative Lag Structures

To examine sensitivity to dynamic adjustment assumptions, we estimate variants of the baseline model with alternative lag lengths: (i) a shorter window (lags 0–1), (ii) a longer window (0–3 and 0–4), (iii) a geometric distributed lag, and (iv) a three-month average tariff index. Magnitude varies modestly across specifications, but the qualitative conclusions remain the same: tariff increases consistently reduce import quantities, with effects accumulating over the first two months.

Tariff-Quantity Decomposition

We further test robustness by decomposing tariff transmission into quantity and price channels. (i) We replace tariff rates with import unit values for dairy and alfalfa, $\Delta \ln P_t^D$ and $\Delta \ln P_t^A$, to isolate price pass-through. (ii) We include both tariff rates and CIF prices jointly in the first stage to test whether tariffs retain predictive power for import quantities after controlling for unit-value movements. Across all decompositions, tariff coefficients remain stable in sign and direction, confirming that results are not artifacts.

Additional Sensitivity Analyses

We also conduct supplemental checks, including excluding the melamine crisis years (2007–2009) and the COVID-19 period (2020–2021), and estimating models in levels rather than log-differences. All robustness exercises yield consistent qualitative patterns, strengthening confidence in the identification strategy.

Table B2: Summary of Robustness Checks

Robustness Exercise	**Description and Expected Outcome**
Placebo Tariff Shocks	Shift tariff series two years earlier; placebo coefficients close zero when tariffs are exogenous.
Lead Tariff Tests	Include future tariff changes; leads have no predictive power for past outcomes.
Short Lag Window (0–1)	Tests immediate adjustment; results align with baseline direction.
Long Lag Window (0–3, 0–4)	Captures delayed contracting/shipping effects; cumulative effects remain similar.
Geometric Distributed Lag	Smooth decay in tariff effects; sign and significance match baseline.
Averaged Tariff Index	Three-month tariff average; stabilizes noise, but effects remain consistent.
Tariff vs. CIF Price Decomposition	Replace tariffs with import prices; tests price pass-through robustness.
Excluding Crisis Years	Removes 2007–2009 (The Melamine Scandal) and 2020–2021 (COVID-19); results remain stable.

C. - Data Description

Tariff data time line

The 2018 U.S.-China trade war introduced a complex phenomenon, in which we define as "twin tariffs"- the simultaneous imposition of duties on upstream production inputs and downstream final products. In July 2018 China imposed an additional 25% retaliatory tariff on U.S. agricultural production include alfalfa, and most U.S. dairy products also became subject to substantial retaliatory duties (Xu et al., 2025). In 2019, many cheese lines faced totals around 33–37%, reflecting China's MFN tariff plus the 25% Section 301 retaliation. China announced another round in September 2019, adding 5% or 10% on selected U.S. goods; for dairy, products such as fresh cheese and grated cheese reached total applied tariffs of about 38–42%, while buttermilk and fermented milk could reach about 50% depending on the line. After the Phase One Agreement was signed in January 2020 and entered into force on February 14, 2020, China did not fully remove retaliatory tariffs, but it opened an exclusion process for affected farm goods and expanded market access for U.S. dairy facilities and forage products, including alfalfa hay pellets and cubes; in practice, this meant many dairy tariffs formally remained, but some products obtained exclusions. From 2021 to 2025, China continued to maintain retaliatory Section 301 tariffs on most U.S. dairy products, although key feed-related exceptions were repeatedly extended; notably, whey for feed use received repeated exclusions through February 28, 2025, and alfalfa had earlier benefited from exclusion treatment beginning in September 2019, which helped ease pressure on China's dairy-feed system even though the broader retaliatory tariff regime largely stayed in place. In May 2025, following the Geneva negotiations, China temporarily reduced the additional retaliatory tariff imposed during the 2025 escalation from 125% to 10% for 90 days. Earlier tariff layers and product-specific measures were not necessarily eliminated in full. So the effective tariff burden on individual agricultural goods could still differ across products. In the Chinese dairy sector, province-level dairy price data are unavailable, but farm-gate milk prices are relatively uniform because production is concentrated in northern provinces and raw milk is primarily purchased by large company such as Mengniu and Yili. In addition, forage tariffs are uniform across the relevant HS6 products, while dairy tariffs vary across HS6 lines. However, most high-variation products, such as butter, lactose, and cheese, represent only a small share of Chinese consumption; the main imported dairy products (milk powder, liquid milk, and whey) face the same tariff rate. Together, these institutional features support the use of the twin-tariff framework as a reasonable sector-level approximation.

Data coding and source

All data process coding are open access on author's GitHub and sole responsibility of the authors: https://github.com/liyoumin/Trade-tarrifs-impacts-on_Chinnese_dairy/blob/main/Trade_twin_tariff.R

Table C1: Data sources and variable descriptions.

Variable	**Source**	**Code**	**Notes**	**Time**
U.S. export dairy value	USDA FAS (GATS)	HS04&0410	$ Thousand	01/2005–06/2025
U.S. export alfalfa value	USDA FAS (GATS)	HS1214	$ Thousand	01/2005–06/2025
U.S. export alfalfa quantity	USDA FAS (GATS)	HS1214	Metric ton	01/2005–06/2025
U.S. export dairy quantity	USDA FAS (GATS)	HS04&0410	Metric ton	01/2005–06/2025
FAO dairy price index	FAO (weighted avg. 2014–2016)	HS04&0410	Index	01/2005–07/2025
China farm-gate milk price	CLAL.it	—	CNY / kg	01/2009–06/2025
China imports forage grass	GACC (China Customs)	HS12	Metric ton	01/2010–06/2024
Feed & fodder quantity	USDA FAS (GATS)	0120AT	Metric ton	01/2010–06/2024
U.S. alfalfa price	USDA NASS	—	$ / ton	2010–2024
U.S. dairy price	BLS via FRED	—	% change	1995–2025
U.S. grazing fee	USDA NASS	—	$ / animal / month	2010–2024
Tariff rate	Chad P. Bown (2025 Chart)	—	—	06/2018–03/2025
Tax revenue (state/local)	U.S. Census Bureau	—	Million $	01/2005–12/2024
Tariff rate (ISIC)	ISIC	—	%	01/2005–12/2024
Ag. export value	USDA ERS	HS04&12	$ Thousand	01/2005–06/2025

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.